

Bayes' Formula Answers

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To compute $P(B_1|A)$ when $P(B_i)$ and $P(A|B_i)$ are given, use the following three-step procedure:

$$P(B_1|A) = \frac{P(B_1 \cap A)}{P(A)}$$

by the definition of conditional probability. Then

$$P(B_i \cap A) = P(B_i)P(A|B_i).$$

Then

$$P(A) = \sum_{i=1}^n P(B_i \cap A).$$

Combining the three steps gives Bayes' formula:

Theorem (Bayes' Formula).

$$P(B_1|A) = \frac{P(B_1)P(A|B_1)}{\sum_i P(B_i)P(A|B_i)}.$$

Exercise 1. Suppose that the number of children in a family is 1, 2, or 3, with probability $1/3$ each. Little Bobby has no brothers. What is the probability he is an only child?

Let B_1 , B_2 , B_3 be the events that a family has one, two, or three children, and let A be the event that a family has only one boy. We want to compute $P(B_1|A)$. By the definition of conditional probability,

$$P(B_1|A) = P(B_1 \cap A)/P(A).$$

To evaluate the numerator we use the multiplication rule

$$P(B_1 \cap A) = P(B_1)P(A|B_1) = \frac{1}{3} \cdot \frac{1}{2} = \frac{1}{6}$$

Similarly, $P(B_2 \cap A) = 1/6$ and

$$P(B_3 \cap A) = P(B_3)P(A|B_3) = \frac{1}{3} \cdot \frac{3}{8} = \frac{1}{8}$$

Now $P(A) = \sum_i P(B_i \cap A) = 1/6 + 1/6 + 1/8 = 11/24$ and

$$P(B_1|A) = \frac{P(B_1 \cap A)}{P(A)} = \frac{4/24}{11/24} = \frac{4}{11}$$

Exercise 2. Little Bobby has no sisters. What is the probability he is an only child?

Answer: Let A be the event that a family has no girls. We have $P(A) = 4/24 + 2/24 + 1/24 = 7/24$. The answer is $P(B_1|A) = 4/7$.

Exercise 3. Binary digits are sent down a noisy communications channel. They are received as sent with probability 0.9 but errors occur with probability 0.1. Assuming that 0's and 1's are equally likely, what is the probability that a 1 was sent given that we received a 1?

Solution: Let B_1 be the event "a 1 was sent" and B_2 be the event "a 2 was sent" and A the event "we received a 1." We want to find $P(B_1|A)$. We have $P(A|B_1) = 0.9$ and $P(B_1) = 0.5$, and $P(B_2) = 0.5$ and $P(A|B_2) = 0.1$. So by Bayes formula,

$$P(B_1|A) = \frac{P(B_1)P(A|B_1)}{P(B_1)P(A|B_1) + P(B_2)P(A|B_2)} = \frac{0.5(0.9)}{0.5(0.9) + 0.5(0.1)} = 0.9.$$

Exercise 4. Two hunters shoot at a deer, which is hit by exactly one bullet. If the first hunter hits his targets with probability 0.3 and the second with probability 0.6, what is the probability the second hunter killed the deer? The answer is not 2/3. Do you think the answer is larger or smaller?

Solution: Let A be the event "the deer is shot with exactly one bullet," B_1 the event that the first and second hunters shot it, B_2 the event the first shot and the second missed, B_3 the event the second shot and the first missed, and B_4 the event that both hunters missed. Then $P(A) = P(B_2) + P(B_3) = .54$ and we want $P(B_3|A) = P(B_3)/(P(B_2) + P(B_3)) = 0.42/0.54 = 7/9$.

Exercise 5. You are about to have an interview for Harvard Law School. 60 percent of the interviewers are conservative and 40 percent are liberal. 50 percent of the conservatives smoke cigars but only 25 percent of the liberals do. Your interviewer lights up a cigar. What is the probability he is a liberal?

Answer: Let A be the event that the interviewer smokes, B_1 the event that he is a liberal, and B_2 the event that he is a conservative. We want $P(B_1|A)$, which is $P(B_1 \cap A)/P(A)$. We have

$$P(B_1 \cap A) = P(A|B_1)P(B_1) = (0.25)(0.4) = 0.1.$$

Also,

$$P(A) = P(A|B_1)P(B_1) + P(A|B_2)P(B_2) = 0.1 + 0.3 = 0.4.$$

So the answer is $0.1/0.4 = 1/4$.

Exercise 6. One slot machine pays off 1/2 of the time, while another pays off 1/4 of the time. We pick one of the machines and play it six times, winning 3 times. What is the probability we are playing the machine that pays off only 1/4 of the time?

Answer: Let B be the event that we are playing the machine that pays off only 1/4 of the time and A be the event that we win 3 out of 6 times. We want to compute $P(B|A)$. By the definition of conditional probability,

$$P(B|A) = P(B \cap A)/P(A).$$

To evaluate the numerator we use the multiplication rule

$$P(B \cap A) = P(B)P(A|B) = (1/2)(20)(1/4)^3(3/4)^3 = 270/4^6.$$

Now $P(A) = P(B \cap A) + P(B^c \cap A)$. We have

$$P(B^c \cap A) = P(B^c)P(A|B^c) = (1/2)(20)(1/2)^6 = 5/32.$$

So

$$P(A) = 270/4^6 + 5/32 = 455/2048$$

So

$$P(B|A) = 135/455 = 27/91.$$

Exercise 7. 20 percent of people are accident-prone and have a probability 0.15 of having an accident in a one-year period in contrast to a probability of 0.05 for the other 80 percent of people. (a) If we pick a person at random, what is the probability he will have an accident this year?

Answer: $0.2(0.15) + 0.8(0.05) = 0.03 + 0.04 = 7$ percent

(b) What is the probability a person is accident-prone if he had an accident last year?

Answer: Let B_1 be the event that a person had an accident last year. Let A be the event that a person is accident-prone. We want $P(A|B_1) = P(A \cap B_1)/P(B_1)$. We know that $P(B_1) = 7/100$. Then $P(A \cap B_1) = P(B_1|A)P(A) = 0.15(0.2) = 0.03$, so the answer is $3/7$.

(c) What is the probability he will have an accident this year if he had one last year?

Answer: Let A be the event that he is accident-prone. We need the probability of A given that he had an accident last year. The probability that he is accident-prone and had an accident last year is $3/7 \cdot 7/100 = 3/100$. The probability that he had an accident last year is $7/100$. So the probability that he is accident prone, given that he had an accident last year, is $3/7$. So we want

$$\frac{3}{7} \cdot \frac{3}{20} + \frac{4}{7} \cdot \frac{5}{100} = 9/140 + 1/35 = 13/140 \approx 0.0929.$$

Exercise 8. Three factories manufacture 20%, 30%, and 50% of the computer chips a company sells. If the probability of defective chips are 0.4%, 0.3%, and 0.2%, respectively, what is the probability that the defective chips come from the third factory?

Answer: Let B be the event that a chip comes from the third factory and A the event that a chip is defective. We want $P(B|A)$. We have $P(B) = 1/2$ and $P(A|B) = 2/1000$, so $P(B \cap A) = 1/1000$. Then for $P(A)$, we get

$$P(A) = 1/5(4/1000) + 3/10(3/1000) + 1/2(2/1000) = 4/5000 + 9/10000 + 1/1000 = 27/10000.$$

So

$$P(B|A) = 10/27.$$

Exercise 9. A company gives a test to 100 salesmen, 80 with good sales records and 20 with poor records. 60 percent of the good salesman pass the test, but only 30 percent of

the poor salesmen do. A new applicant takes the test and passes. What is the probability he is a good salesman?

Answer: Let B be the event that he passes and A the event that he is a good salesman. We want $P(A|B)$, which is equal to $P(A \cap B)/P(B)$. We know that $P(A \cap B) = P(A)P(B|A) = 0.8(0.6) = 0.48$. And $P(B) = 0.8(0.6) + 0.2(0.3) = 0.54$. So the answer is $48/54 = 8/9$.

Exercise 10. Two masked robbers try to rob a crowded bank during the lunch hour but the teller presses a button that sets off an alarm and locks the front door. The robbers, realizing they are trapped, throw away their masks and disappear into the chaotic crowd. Confronted with 40 people claiming they are innocent, the police give everyone a lie detector test. Suppose that guilty people are detected with probability 0.95, and innocent people appear to be guilty with probability 0.01. What is the probability that Mr. Jones is guilty given that the lie detector says he is?

Answer: We have

$$P = \frac{\frac{2}{40} \times .95}{\frac{38}{40} \times .01 + \frac{2}{40} \times .95} = \frac{2 \times 95}{38 + 2 \times 95} = \frac{5}{6}$$